



NuMagSANS Tutorial 2

NuMagSANS Tutorials - Michael P. Adams

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Running the Tutorials

The executable tutorial scripts support two installation modes.

Shared setup: local workstation or HPC shell

`python -m venv .venv`

`source .venv/bin/activate`

Core install: data generation and NuMagSANS runs

`pip install -e .`

Optional local plotting support

`pip install -e ".[tutorials]"`

- Use the core install for batch/HPC runs.
- Missing PyVista only disables diagnostic plots.
- Use the tutorials extra for local vector-field inspection.

Tutorial Goal

- Use this slide deck as a second NuMagSANS tutorial template.
- Replace these placeholder slides with a focused topic.
- Keep shared figures in `docs/_static` where possible.

- Data organization for ensemble simulations.
- StructData and RotData loops.
- Replication import for repeated local objects.

- Local object data describe one or more materialized objects.
- Structure data places these objects in global space.
- Rotation data orients these objects in global space.
- Output data can be exported as CSV or HDF5.

- Add tutorial-specific material here.
- Prefer small, reproducible examples.
- Keep generated files out of version control.